Franklin & Marshall College **PHY 223**

Interferometry 1

This experiment makes it possible for you to perform the famous two-slit interference experiment with light, even in the limit of light intensities so low that you can record the arrival of individual photons at the detector. This experiment has three parts, two of which you will complete this week:

1. You will see two-slit interference visually, by opening up an apparatus and seeing the exact arrangements of light sources and apertures that operate to produce an 'interference pattern'. You'll be able examine every part of the apparatus, and make all the measurements you'll need for theoretical modeling.

2. You will perform the two-slit experiment quantitatively, recreating not only Young's measurement of the wavelength of light, but also getting detailed information about intensities in a two-slit interference pattern that can be compared to predictions of wave theories of light.

3. If time allows in the second week, you will perform the two-slit experiment one photon at a time, continuing the same kind of experiments, but now at a light level so low that you can assure yourself that there is at most one relevant photon in the apparatus at any time. Not only will this familiarize you with single-photon detection technology, it will also show you that however two-slit interference is to be explained, it must be explained in terms that can apply to single photons. (And how can a single photon involve itself with two slits?)

Experimental setup

The apparatus consists of a long rectangular metal assembly, with a single-photon detection box attached at one end (see Fig. 1).



Figure 1: The TeachSpin two-slit interference device. The power supplies and light sources are to the left, the various slits are in the middle, and the photon detectors are on the right.

First (before plugging anything in, or turning anything on) you'll need to confirm that the shutter that protects the amazingly sensitive single-photon detector is closed. Locate the detector box at the right end of the apparatus, and find the rod that projects out of the top of its interface with the long assembly. Be sure that this rod is pushed all the way down; take this opportunity to try pulling it vertically upward by about 2 cm, but then ensure that it's returned to its fully-down position. Also look at the detector box and confirm that the toggle switch in the HIGH-VOLTAGE section is turned off, and that the 10-turn dial near it is set to 0.00, fully counter-clockwise.

Then open the cover and take a look inside. A schematic of the apparatus is shown in Fig. 2. Make sure to identify the following parts:

1. At the left end are two distinct light sources, one a red laser and the other a green- filtered light bulb; also at the left end are the controls for the light sources.

2. Along the length of the long box are the various slits and apertures that form a Young's two-slit experiment. On the front of the box are two 'micrometer drives', which allow you to make mechanical adjustments to the two-slit apparatus.

3. There are two distinct light detectors in the box at the right-hand end of the apparatus:

   - The first detector is called the 'photodiode', which is used with the much brighter laser light source; it's attached to the light shutter in such a way that it's in position for use when the shutter is in its down position.
   - The second detector is called the 'photomultiplier tube' or PMT for short, which is used with the much dimmer light-bulb source. **It is safe to use only when the cover of the apparatus is in place and the light bulb is in use.**

4. Finally, there are the electrical connections. Electric power comes from a power module, plugged into your a.c. line and connected by a cable to the left end of the apparatus. Two more cables run from this end to the detector box and supply the photodiode and PMT detectors with the power they need for operation.

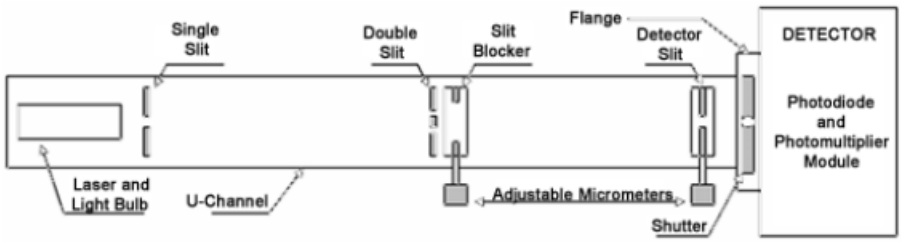


Figure 2: Schematic diagram of the internal arrangement of the TeachSpin two-slit interference apparatus.

Procedure: Visual Mode

For the visual mode of operation, you will be working with the cover of the apparatus open. You'll use the laser as a light source, and you'll find a supply of small white paper screens to be convenient tools. You will not need to use the PMT or photodiode: your eyes are the only detector needed!

Use the controls at the left end of the apparatus to turn on the solid-state diode-laser light source—it's contained within a black metal block. Use a paper screen to find its bright red output beam; the diode laser manufacturer asserts that its output wavelength has a value between 650 nm to 670 nm, and its output power is about 5 mW. (So long as you don't allow the full beam to fall directly into your eye, it presents no safety hazard.) Depending on your experimental setup, you may need to make sure that the laser is centered in the apparatus and that it is not blocked by the lamp for the 'bulb' experiment.

Follow the laser beam until it reaches the entrance slit, or 'source slit', of the two-slit apparatus; this is a single slit of height about 1cm, but width of only about a tenth of a mm. If you apparatus is aligned, the slit will be neatly straddling the laser beam, so that a good fraction of the laser light will be passing through the slit——use your paper screen to see if this is so. If the beam is not aligned with the slit, wiggle and adjust the laser gently until the beam is centered.

Light passing through this narrow source slit will undergo 'single-slit diffraction', and you can follow this process by moving your viewing card downstream. You will see the red beam spread out horizontally, reaching a width of about a cm by the time it reaches the middle of the apparatus.

In the middle of the apparatus is the holder for the double slit, which is a structure with two rectangular apertures, again each about a cm high, and each with width of about a tenth of a mm, and with center-to-center separation of several tenths of a mm. Note that all of the slit structures are mounted on magnetic holders so that they can be removed, examined, and re-positioned; if you apparatus is already aligned (as yours is), it would be a pity to spoil this alignment by disturbing the slits now! Instead, put your viewing card just downstream of the two-slit structure, and look (close up) to see if you can observe the two ribbons of light, a few mm apart, which emerge from the two slits.

This is your chance to understand the function of the slit blocker, just downstream of the double-slit system, which will allow you selectively to block the light coming from either of the two slits. This slit-blocker is controlled by the 'micrometer screw' on the front center of the apparatus; rotating the micrometer's knob will move the slit-blocker laterally across the two ribbons of light emerging from the two-slit system. If necessary, adjust the micrometer until you find a position that permits the two ribbons of light to emerge and continue rightwards in the apparatus.

Each of the narrow ribbons of light emerging from a slit will continue to diffract; follow these broadening ribbons downstream until they visibly overlap. By the time your viewing card reaches the right-hand end of the apparatus, the overlap will be nearly complete, and you'll see that the two overlapping ribbons of light combine to form a pattern of illumination displaying the celebrated 'fringes' named after Thomas Young.

At the far-right end of the apparatus is a final optical element; it's an exit slit, or 'detector slit', of the same size and character as the source slit, except that it's mounted on a moveable structure like the slit-blocker, so it too can have its position adjusted by a micrometer screw drive. The purpose of this detector slit is to allow light from a narrow slice of the interference pattern to pass along to the end of the long apparatus and into the detector box. By translating the detector slit laterally along the interference pattern in space, you can select which part of the pattern will have its light sent on to the detector. Thus by scanning the micrometer screw of the detector slit, you can scan over the interference pattern, eventually mapping out its intensity distribution quantitatively. For now, ensure that the detector slit is located somewhere near the middle of the two-slit interference pattern. Make a note of the value of the micrometer at this position for later reference. *When using the micrometer, note that the first digit shown is in* ***mm*** *and there are additional marks at the half-mm positions.* Thus if the micrometer displays a 6 followed by 34, then the detector slit is at a position of 6.34 mm or 6.84 mm depending on whether it is past the half-way mark.

Before replacing the cover of the assembly, carefully measure the distance between the double-slit and the detector slit. Make note of this distance—you will need it later for your analysis!

Procedure: Quantitative Mode

For this experiment, the box cover needs to be in place, and the shutter of the detector box will still be in its closed, or down, position. This blocks any light from reaching the PMT, but the shutter in its down position correctly centers a 1 cm solid-state photodiode, which acts just like a solar cell: it active generates electric current when it's illuminated. The device is equally sensitive everywhere over its area, so it would measure the total amount of light in over the whole interference patter if it were not for the detector slit. However, with the detector slit at a fixed position, the only light reaching the detector is that from the selected part of the interference pattern; by this means, a single-element, spatially-fixed, large-area detector can serve to record (serially in time) the intensities at various places in the interference pattern. The method of course relies on the fact that the rest of the apparatus is stable in time; happily, the diode-laser source has an output power varying by $< 0.1\%$ in time, and the mechanical stability of the rest of the apparatus is also adequate.

Connect a multimeter to the photodiode output on the detector box. You can test whether or not the voltage you see is due to the light from the laser by toggling the laser on and off. **Make sure to record the value of the voltage when the laser is off.** You will need to subtract that from your

measured values to calculate the effect of the laser.

Once you have successfully confirmed that you can measure the voltage produced by the laser on the photodiode, adjust the position of the detector slit until you find two different maxima in the micrometer, and determine the distance between these maxima.

For your quantitative measurements of the interference, you will want to measure the voltage at regular intervals as you scan the detector slit across the interference pattern. Your measurements should include **at least five** maxima and minima, and you will ideally want **at least 10** data points between each pair of successive maxima. It will also be useful to make sure your range of positions includes the overall maximum in the interference pattern.

You now have all the information you need to take your measurements. Decide on a set of micrometer positions (i.e., a starting position, ending position, and step size) that will enable your final data set to meet the criteria above. Then measure the photodiode voltage at each location. As always in collecting data, make sure to consider the uncertainty on your measurements as well.

Modeling of interference and diffraction

The simplest models for the data you have now taken are based on the assumptions of Fraunhofer diffraction (see appendix below for a review). This theory makes certain assumptions that are not perfectly accurate for a realistic experiment, but we can still use it if they approximately satisfied. For example, the theory assumes light reaches the double-slit assembly as a plane wave, ie. from a source infinitely far away. In reality, your light reaches the double slits from the source slit, about 38 cm away. Whether this makes a noticeable difference depends on the precision of our experiment!

The theory also assumes the double slits are the same width a, a center- to-center separation labeled d, and are infinite. For your apparatus, both a and d are much less than 1 mm, and the slit length is about 10 mm. This is long enough that the effects of having a finite slit length can be ignored.

The theory assumes the light is of a definite wavelength λ; in your experiment, the laser light is monochromatic, with λ between 650 nm and 670 nm.

Theory assumes the light spreads from the double-slit assembly toward a point detector also infinitely far away, so the radiation pattern can be expressed in terms of an angular variable θ, measured in radians from the central axis of the apparatus. Your detector is not point-like, but is effectively 0.085 mm wide (the width of the detector slit). It is also not infinitely far away—hopefully, you measured the actual distance when you had the apparatus open earlier.

The predicted intensity distribution derived from the assumptions of Fraunhofer diffraction is

$$I(\theta) = I_0 \left(\cos\beta\right)^2 \left(\frac{\sin\alpha}{\alpha}\right)^2 \tag{1}$$

where

$$\alpha = \frac{\pi a}{\lambda}\sin\theta \tag{2}$$

$$\beta = \frac{\pi d}{\lambda}\sin\theta \tag{3}$$

and θ is the angle of deflection of the light beam. I_0 is the intensity of the central maximum of the pattern, which depends on the total brightness of the source producing the illumination—note that I_0 controls the overall vertical scaling of the pattern, but it does not affect the shape in any way.

Get some practice graphing this expression, until you understand the role of the two factors in the intensity expression and how each of a, d, and I_0 affect the behavior. Also, get familiar enough with its derivation that you can understand *why* it depends on both a and d, and why there must be (at least) two slits in order to produce this pattern.

There are much more sophisticated theoretical models than this one. In particular, there are discussions of Fresnel-diffraction and sum-over-paths models that avoid some of the approximations made above. Nevertheless, these more sophisticated models do not change any qualitative conclusions you have established, and they make only the slightest quantitative differences to your predictions; the need for deep thought about duality for light remains just as urgent after the models for its behavior have been improved.

Analysis

Use the theoretical model and your measurements to determine a value d, including its uncertainty. There are multiple ways to do this: for full credit, you should use **both** of the methods below and **compare your results** between them.

1. Determine the positions of the minima of the double slit pattern and then plot them as a function of the **order** m (where $m = 1$ for the first minimum, $m = 2$ for the second, etc.). Then fit the slope of this graph to determine the separation of these minima (and its uncertainty), and then compare to the expected separation between successive minima based on your theory. Use this comparison to determine d and its uncertainty.

2. Plot your measurements as a function of *angle*, and also plot the the theoretical equation given above. Determine values of I_0, a, and d that produce the best match between the equation and your data, as well as uncertainties for these values. *The best way to do this would be with a fit, but visual comparison of the data and model may also be helpful.* You may use your estimate for d from method 1 as a starting point for finding good values, and note that you should expect the slit width a to be smaller than the separation d.

Note: in order to compare your data to the theoretical model above, you will need to determine how to convert differences in the positions you measured with the micrometer into angles. Use the geometry of the system as shown in Fig. 3 (including the distance R between the double-slit and the detector slit) to make this determination. It may be useful to use the *small-angle approximation* ($\sin\theta \approx \theta$), but be sure to justify it if so. You will also need to use your data to determine the position that should correspond to $\theta = 0$ (i.e., the precise position of the central maximum).

To submit:

1. A properly formatted graph (axes, labels, error bars, etc) showing of position of the minima as a function of order (m), including the best fit line.

2. A properly formatted graph (axes, labels, error bars, etc) showing of position of the interference pattern with at least 5 fringes and ten data points per fringe. Add a modeling line on top of the data according to Eq. 15.

3. Estimates for I_0, a, and d (including uncertainties) from your graphs above.

4. A brief discussion of your results and uncertainties, including:

    - How do the parameters a and d affect the shape of Eq. 1?
    - How do your two estimates of d compare to each other? Are they consistent? What do you see as the pros and cons of each method?
    - How large are your uncertainties on your final values? Do they seem reasonable? Why or why not? What do you think are the primary sources of random and systematic uncertainty in your measurements, and how could you improve them if you were to repeat the measurement?
    - Does the model of Fraunhofer diffraction seem to fit the data well? Were the assumptions of that model justified in your experiment? How do you know?

Appendix: Fraunhofer diffraction

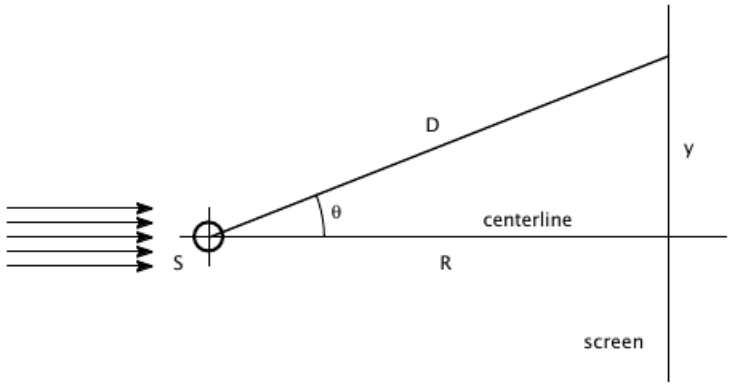


Figure 3: Geometry of Fraunhofer diffraction. An incoming parallel beam is incident on a slit structure S and diffracts to hit a screen at height y above the centerline.

In this appendix we will review how to analyze the single-slit and two-slit interference patterns using the assumptions of Fraunhofer diffraction. The geometric configuration is shown in Fig. 3. The incoming monochromatic light, of wavelength λ, is incident from the left. We assume that the light arrives as a plane wave, which means in practice that the source of the incident light is far to the left. The incoming light is incident on a slit arrangement S, which is assumed to be infinitesimally small compared to the other distances (R, D, y) of the scattering. The slit arrangement S is a source for scattered waves, one of which is shown going at angle θ relative to its original direction. A screen is shown to the right in the diagram at distance R from the scattering center S. The scattered wave travels a distance D before hitting the screen a distance y above the centerline. We will assume that the amplitude of the scattered wave is independent of y, which is saying that D doesn't vary much from R, or that the angle θ is small. The trigonometric relations are

$$\tan\theta = \frac{y}{R} \tag{4}$$

and

$$\sin\theta = \frac{y}{D} \tag{5}$$

Slit arrangements have some structure so that the wave arriving on the screen at y is a sum of contributions coming from the various slit parts. In Fig. 4 a particular wave, coming from an opening located a distance h above the centerline, is shown. This wave moves a distance d, which is slightly different from D, in order to reach the screen a distance y above the centerline. (The vertical dimension is exaggerated in this diagram for clarity.)

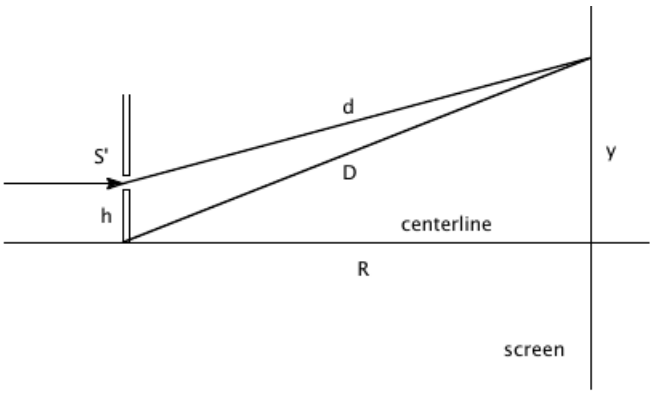


Figure 4: Details of the slit structure in Fraunhofer diffraction. Variations in the wave travel distance d will be important for determining the phase of the wave on the screen.

Waves scattered at the slit structure spread out with a spherical wavefront and are described by the spherical wave $\Psi = \chi e^{i(kd-\omega t)}$, where $k = 2\pi/\lambda$ is the wave number, $\omega = 2\pi f$ is the angular frequency, and χ is the differential amplitude coming from this small contribution to the total scattered wave. We use a complex exponential form to describe the wave, but we could have used a real form by working with the real part of Ψ. The complete wave arriving at any point on the screen is a sum of contributions coming from the various parts of the slit structure. This total contribution is given by the sum

$$\Psi(y) = A \int_{slit} dh e^{i(kd-\omega t)} \tag{6}$$

where $(Adh)e^{i(kd-\omega t)}$ is the small wave contribution coming from a differential length dh of the slit. We have assumed that the quantity A is a constant (independent of h and y, which means we are assuming that the distances between the various parts of the slit structure to the a point on the diffraction pattern on the screen are nearly the same. However, the effects of variations in d in the exponential are significant and we take them into account. We can write

$$d = \left(R^2 + (y-d)^2\right)^{1/2} = \left(R^2 + y^2 - 2yh + h^2\right)^{1/2} \approx D\left(1 - \frac{yh}{D}\right) \tag{7}$$

for small θ (that is, for $y/D << 1$) and small h (that is, for $h/D << 1$) so that

$$\Psi(y) \approx A e^{i(kD-\omega t)} \int_{slit} dh e^{-ikyh/D} \tag{8}$$

We have defined $D = (R^2 + y^2)^{1/2}$. As a first application of this "Fraunhofer" diffraction, we calculate the diffraction pattern formed from a double slit arrangement. We take a slit width a and a slit separation d, so the slit is open for $-(d+a)/2 < h < -(d-a)/2$ and $(d-a)/2 < h < (d+a)/2$. The wave at height y on the screen is

$$\Psi(y) = A e^{i(kD-\omega t)} \left(\int_{-(d+a)/2}^{-(d-a)/2} dh e^{-ikyhD} + \int_{(d-a/2)}^{(d+a)/2} dh e^{-ikyh/D} \right) \tag{9}$$

This integral can be done with the result

$$\Psi(y) = 2aAe^{kD-\omega t} \frac{\sin\alpha}{\alpha} \cos\beta \tag{10}$$

where

$$\alpha = \frac{\pi a}{\lambda} \sin\theta = \frac{kya}{2D} \tag{11}$$

$$\beta = \frac{\pi d}{\lambda} \sin\theta = \frac{kyd}{2D} \tag{12}$$

and we have used the Euler identities

$$e^{i\theta} + e^{-i\theta} = 2\cos\theta \tag{13}$$

and

$$e^{i\theta} - e^{-i\theta} = -2i\sin\theta \tag{14}$$

The intensity is the complex square of the wave amplitude. For the double slit we have

$$I(y) = I_0 \left(\frac{\sin\alpha}{\alpha}\right)^2 (\cos\beta)^2 \tag{15}$$

where $I_0 = 4|aA|^2$ is the maximum of the intensity distribution, which occurs at $y = 0$.

The distribution for a single slit, located at $-a/2 < h < a/2$, is given by

$$\Psi(y) = Ae^{i(kD-\omega t)} \int_{-a/2}^{a/2} dh e^{-ikyd/D} = aAe^{i(kD-\omega t)} \frac{\sin\alpha}{\alpha} \tag{16}$$

with corresponding intensity distribution

$$I(y) = \frac{I_0}{4} \left(\frac{\sin\alpha}{\alpha}\right)^2 \tag{17}$$